

Product Category Management

Tata CliQ E-Commerce Database Management System

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER(5)	Primary Key
Category_Name	VARCHAR2(50)	UNIQUE, NOT NULL

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	NUMBER(5)	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Description	VARCHAR2(200)	—
Price	NUMBER(10,2)	NOT NULL
Stock	NUMBER(5)	NOT NULL
Category_ID	NUMBER(5)	Foreign Key

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key
- Product.Category_ID references Category.Category_ID
- Relationship: One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE CATEGORY (  
2     CATEGORY_ID NUMBER(5) PRIMARY KEY,  
3     CATEGORY_NAME VARCHAR2(50) UNIQUE NOT NULL  
4 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE PRODUCT (  
2     PRODUCT_ID NUMBER(5) PRIMARY KEY,  
3     PRODUCT_NAME VARCHAR2(100) NOT NULL,  
4     DESCRIPTION VARCHAR2(200),  
5     PRICE NUMBER(10,2) NOT NULL,  
6     STOCK NUMBER(5) NOT NULL,  
7     CATEGORY_ID NUMBER(5),  
8     CONSTRAINT FK_PRODUCT_CATEGORY  
9         FOREIGN KEY (CATEGORY_ID)  
10        REFERENCES CATEGORY(CATEGORY_ID)  
11 );
```

Table created.

4. Data Insertion

4.1 Category Records

This table contains the inserted category records.

```
SQL> INSERT INTO CATEGORY VALUES (1, 'Electronics');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (2, 'Mobiles');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (3, 'Laptops');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (4, 'Fashion');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (5, 'Footwear');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (6, 'Beauty');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (7, 'Home Appliances');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (8, 'Watches');

1 row created.

SQL> INSERT INTO CATEGORY VALUES (9, 'Accessories');

1 row created.
```

4.2 Product Records

This table contains the inserted product records.

```
SQL> INSERT INTO PRODUCT VALUES
| 2 (101, 'Samsung Galaxy S24', 'Latest Samsung smartphone', 69999.00, 15, 2);

1 row created.

SQL>
SQL> INSERT INTO PRODUCT VALUES
| 2 (102, 'iPhone 15', 'Apple smartphone with advanced camera', 69999.00, 10, 2);

1 row created.

SQL>
SQL> INSERT INTO PRODUCT VALUES
| 2 (103, 'HP Pavilion Laptop', 'HP laptop for everyday use', 54999.00, 12, 3);

1 row created.

SQL>
SQL> INSERT INTO PRODUCT VALUES
| 2 (104, 'Nike Running Shoes', 'Comfortable running shoes', 3999.00, 25, 5);

1 row created.

SQL>
SQL> INSERT INTO PRODUCT VALUES
| 2 (105, 'Lakme Face Cream', 'Moisturizing face cream', 899.00, 30, 6);

1 row created.

SQL>
SQL> INSERT INTO PRODUCT VALUES
| 2 (106, 'Samsung Smart TV', 'Smart LED television', 45999.00, 8, 1);

1 row created.
```

5. Data Retrieval

5.1 Display Product Records

This table displays all the product records.

```
SQL>
SQL> SELECT * FROM CATEGORY;
```

```
CATEGORY_ID CATEGORY_NAME
```

1	Electronics
2	Mobiles
3	Laptops
4	Fashion
5	Footwear
6	Beauty
7	Home Appliances
8	Watches
9	Accessories
10	Grocery

```
10 rows selected.
```

5.2 Display Category Records

This table displays all the category records.

```
SQL>
SQL> SELECT * FROM PRODUCT;
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

PRICE	STOCK	CATEGORY_ID
-------	-------	-------------

```
-----
```

101		
-----	--	--

Samsung Galaxy S24

Latest Samsung smartphone

69999	15	2
-------	----	---

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

PRICE	STOCK	CATEGORY_ID
-------	-------	-------------

```
-----
```

102		
-----	--	--

iPhone 15

Apple smartphone with advanced camera

69999	10	2
-------	----	---

6. Product Update

This table shows the product record updated from the Product table.

Updated Product Records – Price and Stock Changes

```
SQL>
SQL> UPDATE PRODUCT
  2  SET PRICE = 64999.00,
  3      STOCK = 15
  4  WHERE PRODUCT_ID = 102;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

7. Product Deletion

This shows the product record deleted from the Product table.

Deleted Product Record – Philips Mixer Grinder

```
SQL> DELETE FROM PRODUCT
  2  WHERE PRODUCT_ID = 110;

1 row deleted.

SQL>
SQL> COMMIT;

Commit complete.

SQL>
SQL> SELECT C.CATEGORY_NAME,
  2      P.PRODUCT_NAME,
  3      P.PRICE,
  4      P.STOCK
  5  FROM CATEGORY C
  6  JOIN PRODUCT P
  7  ON C.CATEGORY_ID = P.CATEGORY_ID
  8  ORDER BY C.CATEGORY NAME;
```

8. Category-wise Product Reports

This table displays product details based on their categories.

```
SQL> SELECT C.CATEGORY_NAME,
2      P.PRODUCT_NAME,
3      P.PRICE,
4      P.STOCK
5  FROM CATEGORY C
6  JOIN PRODUCT P
7  ON C.CATEGORY_ID = P.CATEGORY_ID
8  ORDER BY C.CATEGORY_NAME;
```

CATEGORY_NAME
PRODUCT_NAME
PRICE
STOCK
Accessories
HP Laptop Bag
1499
18
Beauty
Lakme Face Cream
899
30
CATEGORY_NAME
PRODUCT_NAME
PRICE
STOCK

8.1 Mobile Category Product Report

Mobile Category Product Details

This table displays the products under the Mobiles category.

Mobiles
iPhone 15
64999
15
CATEGORY_NAME
PRODUCT_NAME
PRICE
STOCK
Mobiles
Samsung Galaxy S24
69999
15

8.2 Category-wise Total Product Count

This table displays the total number of products in each category.

Category-wise Total Product Count

```
SQL> SELECT C.CATEGORY_NAME,  
2      COUNT(P.PRODUCT_ID) AS PRODUCT_COUNT  
3 FROM CATEGORY C  
4 LEFT JOIN PRODUCT P  
5 ON C.CATEGORY_ID = P.CATEGORY_ID  
6 GROUP BY C.CATEGORY_NAME  
7 ORDER BY C.CATEGORY_NAME;
```

CATEGORY_NAME	PRODUCT_COUNT
Accessories	1
Beauty	1
Electronics	1
Fashion	1
Footwear	1
Grocery	0
Home Appliances	0
Laptops	1
Mobiles	2
Watches	1

10 rows selected.

SQL>